

OFFLINE TRIAGE · V0.1

Sentinel *Health*

Multimodal Gemma 4, a deterministic safety net, and a WhatsApp handoff to the hub physician.

Runs entirely on a clinic laptop with no internet. Scoped to the five emergencies where rural CHWs lose patients to delay: trauma, poisoning, snake bite, MI, stroke.

GEMMA 4 GOOD HACKATHON

FIVE TAI-VADE EMERGENCIES

EN · HI · TA · ML

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What an ASHA worker actually has at 9:30 PM on a Tuesday

THE CHW IN THE VILLAGE

"This patient looks serious. Should I send her to the PHC, or wait?"

Today: an MBBS-trained physician is 38 km away. No textbook in the bag.

THE HUB PHYSICIAN ON WHATSAPP

"Send me the patient's symptoms — but write it the way I need to read it."

Today: a voice note of vague terms. No vitals format, no escalation tier.

THE DISTRICT HEALTH OFFICE

"Which CHW handled what, in which village, with what outcome?"

Today: a paper register. No record-of-care, no analytics.

India has roughly one million ASHAs. The blocker on rural emergency outcomes isn't medicine. It's the 4–6 hour gap between the first physical contact and the first physician opinion.

A real case from the demo · 32 seconds, end-to-end

60-year-old woman. Jaw pain, nausea, fatigue, one hour. Diabetic + hypertensive. The CHW types it in Tamil.

t=0s	CHW opens Sentinel on the offline laptop. Picks the Tamil chip. Types or speaks the symptoms.
t=2ms	Deterministic safety net fires: female · jaw pain · diabetic match the atypical-MI keyword set. <code>force_red=true</code> regardless of what Gemma says next.
t=4s	Gemma 4 reads the symptoms, picks STEMI/NSTEMI from the KB candidates. JSON Schema rejects any free-text invention.
t=10s	CHW taps the camera, attaches the 12-lead ECG photo. Gemma 4's vision module confirms ST elevation in leads II, III, aVF.
t=24s	• RED · differential, during-transport protocol, thrombolysis criteria. WhatsApp handoff message generated in the PHC group format with ambulance number + ETA.
t=32s	CHW taps "Send to hub physician" — the message opens in WhatsApp, pre-typed. Audit record + ECG photo persist locally.

The safety net cannot be talked out of `RED` by a more "reassuring" LLM. Gemma's job is the structured plan, not the triage class.

The four-stage pipeline

01

Safety net

< 5 ms

Keyword red-flag overrides. ~38 atypical-MI / stroke / poison patterns. Curated with Dr. Hari, not scraped.

02

Symptom intake

voice or text

Web Speech API for voice (Chrome cloud STT, Apple on-device STT). 4 languages. Optional ECG/wound photo.

03

Gemma 4 grounded

3-5 s GPU · 20-40 s CPU

`gemma4:e4b-it-q4_K_M` via Ollama. JSON Schema response. Multimodal: text + ECG/wound photo.

04

WhatsApp handoff

< 100 ms

`wa.me` deep-link in real PHC group format. Ambulance, ETA, during-transport protocol.

The safety net runs first because the model can be wrong, but the safety net is built from a clinician's red-flag list. Gemma's structured plan never determines triage class on its own.

Why Gemma 4 — and where it specifically earns its place

MULTIMODAL IN 8B / SUB-4B INFERENCE

Gemma 4 reads ECG photos, wound images, and pill bottles. At `q4_K_M` the model is ~9.6 GB on disk with a sub-4B inference footprint — small enough to bake into a Cloud Run image and small enough to load on a 16 GB laptop alongside Ollama.

JSON SCHEMA-ENFORCED OUTPUT VIA OLLAMA

We pass an explicit `format` schema to Ollama. The model returns `{triage, condition_id, rationale, transport}` — it can't free-form into "this might be MI but possibly anxiety". `condition_id` is constrained to the curated KB list, so it picks, never invents.

OPEN WEIGHTS · NO API KEY · NO PER-CALL COST

Every other comparable clinical AI assumes a working internet and a credit card. Sentinel doesn't. A state health system can deploy this on refurbished laptops, no recurring spend. We tested on three models (Gemma 4 8B, MedGemma 4B Q8, MedGemma 4B) — **100% sensitivity held across all three.** The safety net carries; the model is replaceable.

4-LANGUAGE REASONING, WITH AN HONEST STT CAVEAT

Gemma reasons in English, Hindi, Tamil, Malayalam. Voice transcription is a separate question: we tried Whisper-small for offline STT and Indian-language quality wasn't acceptable. The shipping app uses Chrome's cloud STT

Live demo · 5:50 walkthrough

t	Beat	What you see
0:00	Empty state · Tamil selected · five scope chips	How-to · TAI-VADE chips · example cards
0:30	Cardiac · RED · 60yo woman, jaw pain, diabetic	• RED · safety net cited · STEMI/NSTEMI
1:40	ECG attach · camera button on the offline laptop	Gemma vision · ST elevation called · differential refined
2:50	Snake bite · RED · monsoon season, rural India	• RED · neurotoxic protocol · ASV dose
3:40	Found unconscious · no history	Photograph pills/bottles · partial input handling
4:30	WhatsApp handoff · pre-typed message opens	PHC group format · ambulance # · ETA
5:20	Audit log · the village's record-of-care	Append-only JSONL · ECG side-file persists

Two demo URLs: triage.accurateai.org/demo (full PHI features, runs on my laptop) and huggingface.co/spaces/sankara68/sentinel-health (CPU, always-on, PHI persistence off).

What's measured — and where the safety net earns its keep

100%

SENSITIVITY • 29 / 29 RED CASES

Zero missed red flags across the evaluation set. **True across all three models tested** (Gemma 4 8B, MedGemma 4B Q8, MedGemma 4B). The safety net, not the model, carries this.

80%

SPECIFICITY • 29 / 31 OVERALL

2 GREEN cases triaged YELLOW by an over-cautious model. Acceptable failure mode — the cost of a false escalation is a phone call; the cost of a missed MI is irreversible.

197

PYTESTS PASSING

Unit + integration. Covers the safety net keyword rules, JSON-Schema validation, the WhatsApp message generator, image validation, audit-log roundtrip.

4

LANGUAGES • INDIC SCRIPTS

English, Hindi, Tamil, Malayalam — UI, voice intake, Gemma reasoning prompts, escalation message. Reviewed end-to-end with Dr. Hari.

The model is a writer. The safety net is the doctor. That's the originality.

Three buyers · one deployment unit

ASHA / CHW

Before: guess and refer. No textbook, no second opinion until the PHC.

After: 30-second structured triage, in her own language, with a ready-to-send escalation.

HUB PHYSICIAN (PHC)

Before: WhatsApp voice notes she has to re-interpret while seeing other patients.

After: messages in her own group format with vitals, differential, ETA. Reads in 5 seconds.

DISTRICT / NHM

Before: paper register. No outcome tracking.

After: append-only audit log per laptop. De-identifiable rollup for the district health office.

~1M

ASHAS · INDIA ALONE

National Health Mission workforce. **Add Bangladesh, Indonesia, Sub-Saharan Africa** — the CHW model is the standard low-resource primary care interface.

\$0

RECURRING · PER CHW

Open weights, runs on refurbished hardware. No subscription, no per-call API fee, no PHI on a third-party server. State health systems can budget this.

~5 hrs

TIME-TO-PHYSICIAN GAP

Field-typical delay from first symptom to first physician opinion in remote Tamil Nadu (Dr. Hari's observation). Sentinel target: **under 5 minutes**.

What's next — and what we deliberately left out

OFFLINE STT DONE RIGHT

Whisper-small was shipped and reverted: English fine, Tamil/Malayalam unacceptable. Path forward is either a fine-tuned Whisper on Indic medical terms or AI4Bharat's IndicConformer — not a stock open model. Until then, voice uses Chrome STT with a clear connectivity hint.

HUB PHYSICIAN DASHBOARD

Today the handoff lands in a WhatsApp group. Next: a lightweight dashboard at the PHC tablet for the physician to triage incoming messages, acknowledge them, and close the loop back to the CHW.

SCOPE EXPANSION BEYOND TAI-VADE

Obstetric emergencies (post-partum hemorrhage, eclampsia), pediatric severe fever, snake-bite serotyping by photo. Each addition is a KB entry + a safety-net keyword set + a few eval cases. The architecture doesn't change.

WHAT WE LEFT OUT ON PURPOSE

No generic "ask the AI anything" mode. No free-form chat. No conditions outside the five-emergency scope. The safety net only works because the scope is bounded; broadening the surface would dilute the only thing that makes this safe.

TRY IT · < 30 SECONDS

Sentinel *Health*

One laptop, no internet, four languages, five emergencies.

Live demo on the clinic laptop via Cloudflare tunnel. Always-on public demo on Hugging Face Spaces (CPU, ~30 s/call).

```
# Live demo + project pages
open https://triage.accurateai.org/demo
open https://sentinel.accurateai.org
open https://huggingface.co/spaces/sankara68/sentinel-health
# Or clone & run yourself
git clone github.com/SankarSubbayya/sentinel-health
ollama pull gemma4:e4b-it-q4_K_M
uv sync && uv run uvicorn main:app --port 8000
open http://localhost:8000/demo
```